Neuroscience - Structural Correspondence with the 5-Stage Model

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Neuroscience

Survey of structural correspondence with the 5-stage model (Field · Wave · Edge · Vortex · Bundle)

Survey Overview

- **Scope**: 11 major theories in neuroscience
- **Research question**: Do neuroscientific theories correspond structurally to the 5-stage model?
- **Results**: Strong correspondence in 10 cases, conditional correspondence in 1 case
- **Temperature distribution**: Approximately 60% near-confirmed findings, 30% well-supported findings, 10% at hypothesis stage
- This domain showed one of the densest structural correspondences among all 30 domains surveyed

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Overview of the 5-Stage Model

Stage	Definition
Field (ba)	Undifferentiated state. Initial condition in which neither direction nor structure has yet been determined
Wave (nami)	Exploration stage in which multiple directions diverge and compete
Edge (en)	Tension state in which opposing elements coexist without converging on either side. The place where things meet at a boundary, influence each other, and relationships are formed
Vortex (uzu)	Stage in which new coherence (order) spontaneously emerges from tension
Bundle (taba)	Stage in which form is fixed and stabilized as a reusable structure

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Figure 1: Domain — Research Overview Infographic

Overview of Structural Correspondences

#	Theory / Concept	Key Researchers	Corresponding Stages	Judgment
1	Predictive coding / Free energy principle	Friston, Clark, Rao & Ballard	Field · Wave · Edge · Vortex · Bundle	Strong
2	Interoception and interoceptive inference	Craig, Seth, Barrett	Field · Wave · Edge · Vortex · Bundle	Strong
3	Constructed emotion theory	Barrett, Damasio, LeDoux	Field · Wave · Vortex · Bundle	Strong
4	Executive function and cognitive control	Miller & Cohen, Botvinick, Aron	Wave · Edge · Vortex · Bundle	Strong
5	Consciousness and global workspace	Dehaene, Tononi, Seth	Cross-cutting (threshold condition)	Strong
6	Sleep, dreams, and the prediction-error loop	Hobson, Nir & Tononi, Domhoff	Field · Wave · Vortex	Strong
7	Polyvagal theory	Porges	Field · Wave · Edge · Vortex · Bundle	Strong
8	Embodied cognition / Enactivism	Varela, Thompson, Maturana	All stages (cross-cutting)	Strong
9	Synaptic plasticity / STDP	Bliss & Lomo, Markram, Bi & Poo	Field · Wave · Edge · Vortex · Bundle	Strong
10	Neural oscillations and coherence	Fries, Canolty	Field · Wave · Edge · Vortex · Bundle	Strong
11	Neural avalanches and criticality	Beggs & Plenz	Field · Wave · Edge · Vortex · Bundle	Conditional

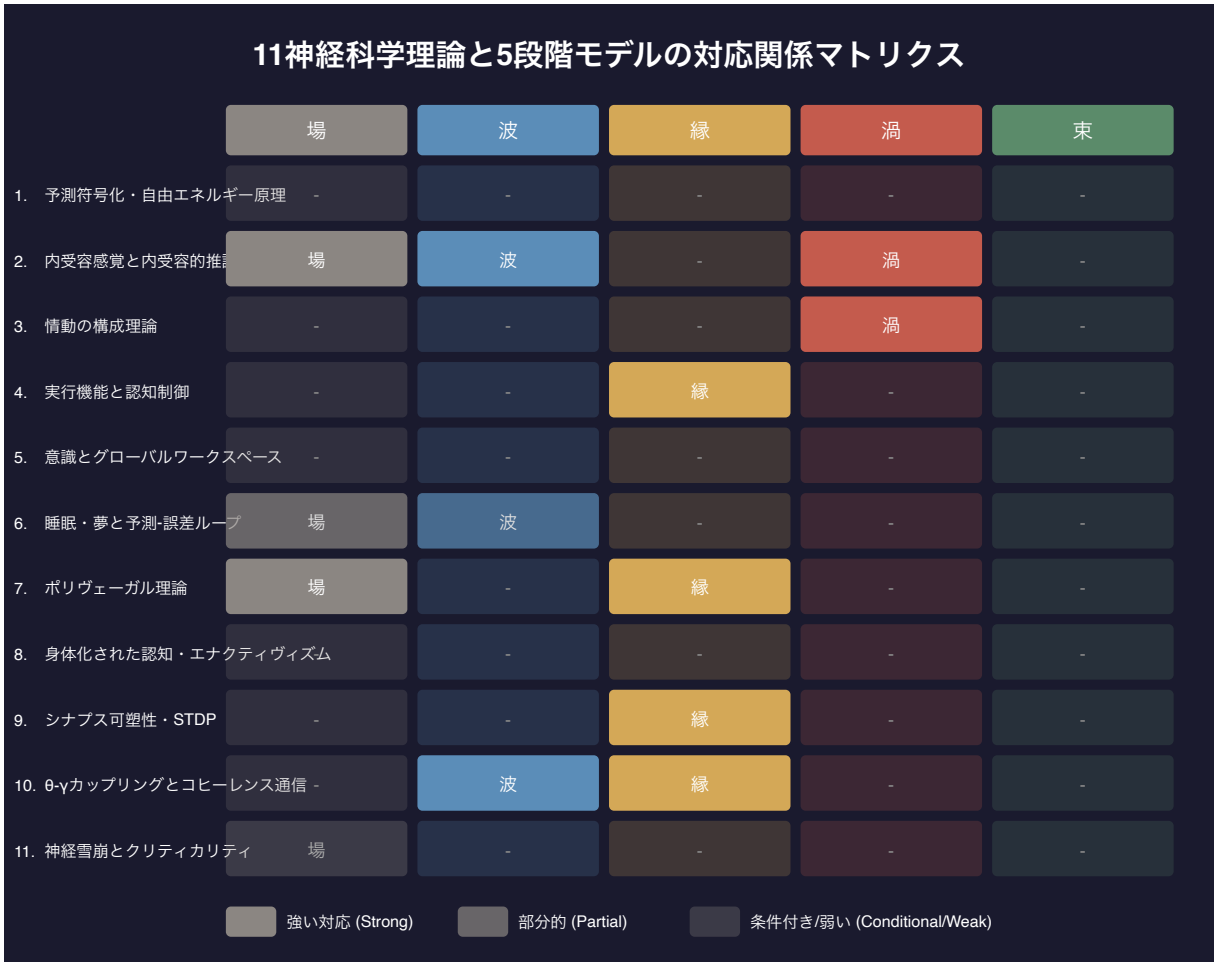

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

Key Entry 1: Predictive Coding and the Free Energy Principle

- **Summary**: The brain does not passively wait for sensory input; it continuously predicts what input will arrive next. Higher layers send predictions downward, while lower layers return prediction errors — a bidirectional process (Rao & Ballard 1999, Friston 2010)
- **Structural correspondence**: All 5 stages
  - The existing predictive model as a whole corresponds to the **Field**
  - The mismatch between predictions and sensory input corresponds to the **Wave**
  - The precision-weighted filtering of which errors to register corresponds to the **Edge**
  - Revision of the existing model corresponds to the **Vortex**
  - Stabilization of the new predictive model corresponds to the **Bundle**
- **Note**: Reading precision as Edge remains open to debate. Precision functions primarily as a control parameter, and there is some distance between that characterization and the Edge description of "meeting at a boundary and mutually influencing each other"

Key Entry 2: Synaptic Plasticity and STDP

- **Summary**: The difference in firing timing (tens of milliseconds) between pre- and postsynaptic neurons determines whether synaptic connections are potentiated or depressed (Markram et al. 1997, Bi & Poo 1998)
- **Structural correspondence**: All 5 stages
  - The existing distribution of synaptic connections corresponds to the **Field**
  - The spike train as a temporal event corresponds to the **Wave**
  - The relative timing of pre- and postsynaptic firing corresponds to the **Edge** (a relational quantity that determines direction)
  - The self-reinforcing loop through repetition corresponds to the **Vortex**
  - The stable connection pattern as a trace of learning corresponds to the **Bundle**
- **Note**: This process occurs without consciousness. Structural change is directed solely by a physical quantity — the time difference — making this an important case demonstrating that the 5-stage structure can operate even at levels that do not presuppose consciousness

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Key Entry 3: Polyvagal Theory

- **Summary**: The autonomic nervous system comprises three hierarchical systems (dorsal vagal system, sympathetic nervous system, ventral vagal system), each supporting different behavioral strategies (Porges 2011)
- **Structural correspondence**: All 5 stages
  - The safe relational field established by activation of the ventral vagal system corresponds to the **Field**
  - Sympathetic activation and bodily arousal correspond to the **Wave**
  - Autonomic regulation at the boundary with another person corresponds to the **Edge**
  - Synchrony and co-regulation between the autonomic nervous systems of two individuals corresponds to the **Vortex**
  - The establishment of a relationship as a secure base corresponds to the **Bundle**
- **Note**: This demonstrates that the Edge operates at the bodily boundary with another person. The events described take place between bodies, mediated by the physiological mechanism of the autonomic nervous system

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Key Entry 4: Embodied Cognition and Enactivism

- **Summary**: Cognition is not the product of the brain alone; it arises from the interaction of brain, body, and environment (Varela, Thompson & Rosch 1991). Autopoiesis refers to the organizational pattern by which life generates and maintains itself
- **Structural correspondence**: Cross-cutting across all stages
  - The coupled system of body and environment corresponds to the **Field**

  - Sensorimotor exploration corresponds to the **Wave**
  - The autopoietic "membrane" (the boundary between body and environment) corresponds to the **Edge**
  - The generation of meaning through action corresponds to the **Vortex**
  - Stabilization as habit or skill corresponds to the **Bundle**
- **Note**: The structure of "meeting at a boundary and mutually influencing each other" repeats isomorphically at different scales — from the cell membrane to the body, from the body to the environment. This supports the possibility that the 5-stage model is a scale-independent structural description

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Structural Correspondence in Detail

Stage	Correspondence in This Domain	Representative Theories / Concepts
Field	Existing predictive model, safe relational field, background rhythm, connection distribution	Predictive coding, Polyvagal theory, CTC, STDP
Wave	Generation of prediction error, arousal and exploration, spike trains, oscillations	Predictive coding, Polyvagal theory, STDP, CTC
Edge	Precision-weighted filtering, direction determined by time difference, phase synchrony, autonomic co-regulation with others	Predictive coding, STDP, CTC, Polyvagal theory
Vortex	Model revision, self-reinforcing loop, phase-locking entrainment, co-regulation	Predictive coding, STDP, CTC, Polyvagal theory
Bundle	New predictive model, stable connection pattern, establishment of relationship	Predictive coding, STDP, Polyvagal theory

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Cross-Cutting Patterns

- **"Relationship determines direction" structure**: In STDP, the time difference in firing; in CTC, phase synchrony; in polyvagal theory, autonomic entrainment — each uses a different physical quantity to select the direction of structural change. The Edge is not a specific mechanism but a structural feature: "selection through relationship"
- **Scale invariance**: The pattern of "detection of difference, formation of relationship, structural change" repeats isomorphically from the synapse (milliseconds) to interpersonal relationships (years)
- **Relationship to consciousness**: STDP executes structural change without consciousness. The 5-stage structure operates regardless of whether consciousness is present; consciousness is positioned as part of the process, not as a prerequisite
- **Field as precondition**: The Field is not a blank starting point but a state in which the conditions for receiving change are in place. Safety (polyvagal theory) and proximity to criticality (neural avalanches) correspond to those conditions

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Open Questions

- How to formally characterize the structural feature common to multiple implementations of the Edge (time difference, phase synchrony, social engagement)
- The validity of reading precision as Edge — the tension between its character as a control parameter and the Edge description
- Establishing the independence of neural avalanches and criticality from complexity science (SOC)
- Integrating criticisms of the neuroanatomical details of polyvagal theory (e.g., Grossman & Taylor 2007)
- The isomorphism of the 5-stage structure across levels with and without consciousness is well-supported but still at the hypothesis stage

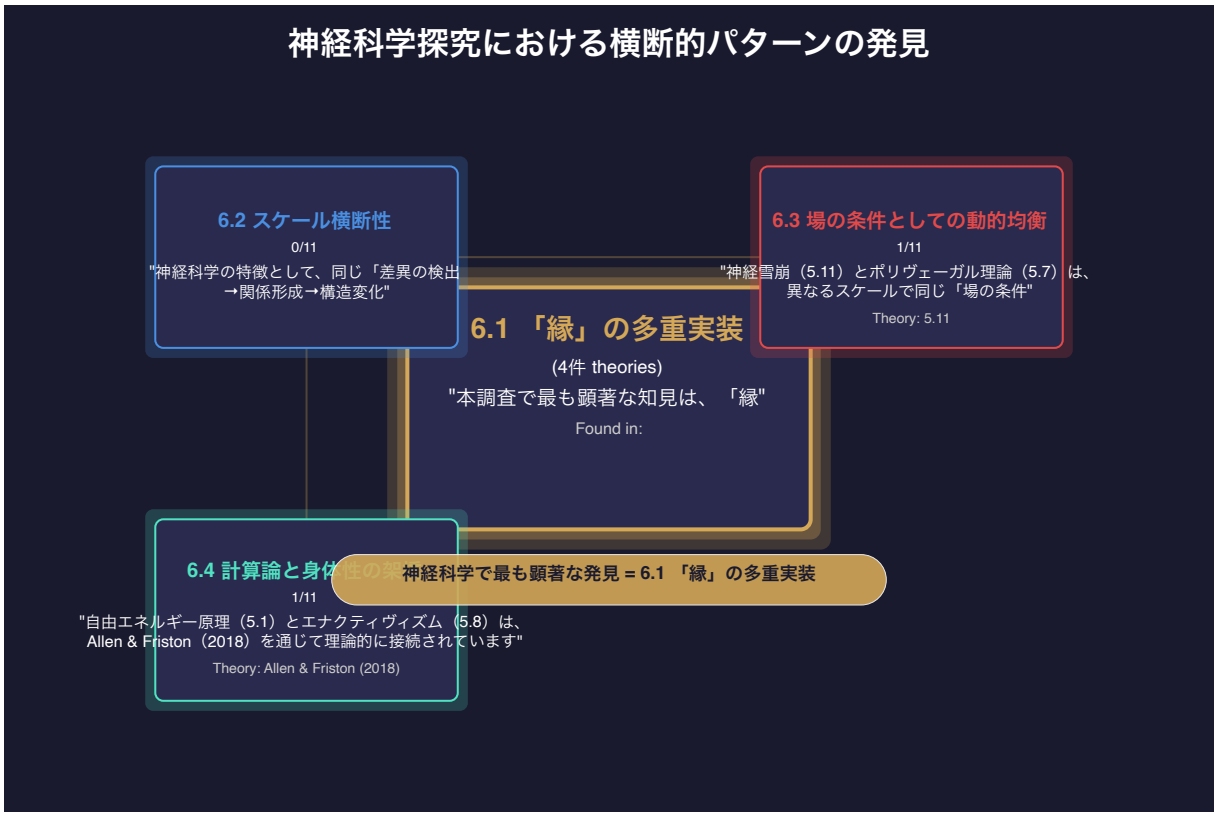

Figure 3: Domain — Cross-Cutting Patterns Diagram

Conclusion

- Neuroscience is one of the domains with the densest structural correspondence to the 5-stage model among all 30 domains. Strong correspondence was confirmed in 10 out of 11 cases
- **Most noteworthy findings**:
  1. The "relationship determines direction" structure appears repeatedly across different scales — from the synapse to interpersonal relationships
  2. The 5-stage structure does not presuppose consciousness. Synaptic plasticity executes structural change without consciousness
  3. The Field is not a blank starting point but a state in which the conditions for receiving change are in place
- **Temperature disclosure**: Near-confirmed findings (predictive coding, synaptic plasticity, etc.) account for approximately 60%; well-supported but still-being-verified findings (causal relationships in CTC, details of polyvagal theory, etc.) for approximately 30%; hypothesis-stage findings for approximately 10%